



Effects of carbon dioxide and green space on sleep quality of the elderly in rural areas of Anhui Province, China

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Abstract

Sleep quality leads to many healthy problems, which has caused global concern. Most studies have focused on the sleep quality in some large cities, ignoring the elderly in rural areas who may have more serious sleep problems. Therefore, this study aimed to understand the sleep status of the elderly in rural areas of Anhui Province and the influence of environmental factors on it. The data of carbon dioxide (CO₂) concentrations for this study were obtained from the Dalhousie University atmospheric composition analysis group. The data of normalized differential vegetation index (NDVI) in 2014 was produced and processed by remote sensing inversion on the basis of medium resolution satellite images. The 2686 individuals were selected from rural areas of Anhui Province by multi-stage stratified cluster sampling. The Pittsburgh Sleep Quality Index (PSQI) was used to assess sleep quality. The independent, interactive, and mediating effects of CO₂ and green space on sleep quality were evaluated by generalized linear model, interaction model, and mediating effect model, respectively. In this study, we found that the prevalence of sleep disturbance in the elderly was 58.40%. In the univariate model, we found that the risk of sleep disturbance increased by 1.6% for each 1 µg/m³ increase in concentrations of CO₂, while decreased by 1.5% for every 0.1 increase in NDVI. In the interaction model, the negative correlation between green space and sleep quality decreased with concentrations of CO₂ increased. In addition, green space was a mediating factor between CO₂ and sleep quality in the mediating effect model.

Keywords Carbon dioxide · Green space · Sleep quality · Rural area · Mediation analysis · Cross-sectional study

Introduction

Sleep quality, as an important factor affecting public health, has attracted worldwide attention. Many studies have shown that sleep quality had a direct impact not only on common chronic diseases such as cardiovascular disease, diabetes, obesity influenza, and mental health (Shankar et al 2010; Qureshi et al 1997; Tarokh et al 2016; Prather et al 2015), but also played an important role in cognitive, chronic pain, digestive, and respiratory disease in older people (Yaffe et al 2011; Andre et al. 2020; Yaremchuk 2018). A literature review showed that sleep quality in the elderly was a risk factor for Alzheimer's disease and was associated with cognitive decline (André et al 2021). Meanwhile, another review has shown a higher prevalence of sarcopenia in sleep-deprived older adults (Rubio-Arias et al. 2019). Many epidemiologists were looking at factors that affected sleep quality, such as age, gender, income level, stress, marital status, chronic

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Highlights

- The elderly in rural areas of Anhui Province had serious sleep problems.
- There was a positive correlation between CO₂ and sleep quality.
- Green space was negatively associated with sleep quality.
- CO₂ could attenuate effect of green space on sleep quality.
- Green space was a mediating factor in the effect of CO₂ on sleep quality.

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disease, and environmental factors (Smyth et al 2020; Dong et al 2018; Hartz et al 2007; Wang et al 2019).

Carbon dioxide (CO₂), as a common greenhouse gas, mainly came from industrial emissions and automobile exhaust. The effects of carbon dioxide on the respiratory system, gut flora, kidneys, blood–brain barrier, and human sleep have been documented in many studies (Khouri et al 2008; Kennedy 1960; CROSS et al 1953; Juan et al 1984). A UK review suggested that prolonged exposure to high levels of CO₂ could cause damage to the muscle tissue, the visual system, and the respiratory system (Colasanti et al 2012). A Chinese experimental study has showed a strong correlation between CO₂ and sleep quality (Xu et al 2021). The Dutch study also found that CO₂ had an effect on sleep efficiency and the number of wakes during the night (Mishra et al 2018). The effects of green space on health have been widely discussed by researchers globally, including the health effects of green space on mental health, as well as reduced risks of preterm birth, type 2 diabetes, stroke, high blood pressure, dyslipidemia, asthma, and coronary heart disease (Moreira et al 2020; Dadvand et al 2016; Twohig-Bennett and Jones 2018; Xiao et al 2020). A review about China has linked green space with mental health, healthier weight status, and better cardiometabolic and cerebrovascular outcomes (Rahimi-Ardabili et al 2021). Another review of children found that exposure to green space was associated with increased levels of physical activity and reduced risk of neurodevelopmental problems such as obesity and inattention in children (Islam et al 2020). In addition, studies on green space and sleep in the America and Australia have found that green space has a protective effect on sleep quality (Astell-Burt et al 2013; Johnson et al 2018).

China, as one of the most populous countries in the world, the insomnia rate was 15.0%, while the rate for people over 43.7 years old was 11.6% according to a meta-analysis study (Cao et al 2017). The China Sleep Quality Index 2017 has reported that the average sleep time of Chinese adults was 6.5 h, while that of Canadian adults was 7.1 h (Chaput et al 2017). Anhui Province is one of the regions where Chinese economic plates cross. A recent study showed the insomnia rate was 24% in older population of Anhui Province, which was higher than the national rate (Ma et al 2018). In recent decades, rapid industrialization and economic development in Anhui Province have accelerated the pace of people's lives, which may lead to higher carbon emissions and sleep problems. However, there were no studies to explore the relationship between CO₂ and sleep quality in rural older population of Anhui Province at present, as well as the interaction and mediating effect of CO₂ and green space on sleep quality. Based on the cross-sectional study focusing on the elderly in Anhui Province, we aimed to explore the

combined effects of CO₂ and green space on the elderly in rural older population of Anhui Province.

Material and methods

Study area

Anhui Province is located in the east of China, the lower reaches of the Yangtze River and the middle reaches of the Huaihe River, between 114° 54'–119° 37' east longitude and 29° 41'–34° 38' north latitude. It covers an area of 140,100 km² and has a permanent population of 63.659 million. Anhui Province is in the transition zone between warm temperate zone and subtropical zone, with four distinct seasons. The north of the Huaihe River has a warm temperate semi-humid monsoon climate, while the south of the Huaihe River has a subtropical humid monsoon climate, with an annual average temperature of 14–17 °C and an average precipitation of 773–1670 mm.

Study population

Definition of study population

In 2006, the World Health Organization (WHO) in its report on the health of the global population recommended that the age limit for older people should be set according to the socio-economic context of each country, with 60 years or more for developing countries. Therefore, the age of 60 years or more was used as the cut-off point for the elderly in this study (Ohayon et al 2004).

Inclusion and exclusion criteria of study population

Inclusion criteria are as follows: (1) residents aged 60 or above in the survey area and (2) the registered population who have lived in the village for more than half a year.

Exclusion criteria are as follows: (1) Alzheimer's patients, (2) older people with severe visual and hearing impairments, and (3) older people who suffer from other serious diseases and cannot communicate normally.

Sampling principles and methods

In order to ensure the representativeness and scientificity of sampling, a multi-stage stratified cluster sampling method was adopted. In the first stage, Anhui Province was divided into three regions according to its geographical location and economic status: northern Anhui, central Anhui, and southern Anhui, and two cities or counties were selected from each region from June 2014 to September 2014. In the second stage, 8, 7, and 4 administrative villages were

randomly selected from the cities or counties in each region, and then, the face-to-face questionnaires survey were conducted among 2686 individuals from 19 administrative villages. Finally, 2642 valid questionnaires were obtained by eliminating unqualified questionnaires in 5 counties or districts of Anhui Province, with the qualified rate of 98.4%.

Sleep quality assessment

Sleep quality was assessed by the Pittsburgh Sleep Quality Index (PSQI), which was a self-rating sleep quality scale developed by the Sleep and Biological Rhythm Research Center, Department of Psychiatry, University of Pittsburgh Medical Center (Buysse et al 1989). It contains of 19 self-rated items (consisting of 9 self-rated questions) and 5 other-rated items, of which 19 self-rated items comprise 7 sleep factors: sleep quality, sleep duration, sleep duration, sleep efficiency, sleep disorders, hypnotic drugs, and daytime dysfunction. Each component was scored on a 0–3 scale, and the cumulative score of all component were the total score of PSQI. The overall score ranged from 0 to 21, with the higher the score, the worse the sleep quality. The Chinese version of PSQI had good representativeness and repeatability through large sample studies (Tang et al 2017). The total score of PSQI > 7 is defined as having sleep disturbance, and the total score ≤ 7 is defined as having good sleep (Buysse et al 1989; Li et al 2013).

Exposure assessment

CO₂ assessment

The data of carbon dioxide (CO₂) concentrations for five districts and counties were obtained from the Dalhousie University atmospheric composition analysis group (http://fizz.phys.dal.ca/~atmos/martin/?Page_id=140). The specific data processing steps are as follows: they estimated ground-level CO₂ total and compositional mass concentrations over China by combining aerosol optical depth (AOD) retrievals from the NASA MODIS, MISR, and SeaWiFS instruments with the GEOS-Chem chemical transport model, and subsequently calibrated to regional ground-based observations of both total and compositional mass using geographically weighted regression (GWR).

Green space assessment

In this study, the normalized differential vegetation index (NDVI) was used as an evaluation index of green space. We estimated the annual average exposure value of NDVI according to the area code corresponding to the specific address of the survey population. The NDVI distribution data product of Anhui Province in 2014 was produced

and processed by remote sensing inversion on the basis of medium resolution satellite images, with a spatial resolution of 250 m. In addition, we combined cloud-free images from the summer of 2014 to create a map covering the entire province of Anhui, from which we picked out maps of green space density for the 5 districts or counties we studied (Fig. 1).

NDVI mainly reflects the difference between the reflection of vegetation in visible and near-infrared bands and the soil background. Each vegetation index can be used to quantitatively explain the growth status of vegetation under certain conditions, and can detect the growth status of vegetation and vegetation coverage and eliminate some radiation errors. The value range of NDVI is between −1 and 1. Negative value means that the ground cover is cloud, water, snow, etc., with high reflection to visible light; 0 indicates rock or bare soil, etc.; positive value indicates the presence of vegetation coverage, and increases with the increase of coverage. NDVI can reflect the background effects of plant canopy, such as soil, wet ground, snow, dead leaves, and roughness, and was related to vegetation cover. We set the negative value of NDVI to 0 (Klompaker et al 2019). The calculation equation of NDVI is as follows:

$$NDVI = (NIR - R) / (NIR + R)$$

NIR is the pixel value of the infrared band, and *R* is the pixel value of the red band.

All the calculation of NDVI and the processing of geographical distribution were carried out on ArcGIS 10.2 software (v.10.2, ESRI, Redlands, CA, USA).

Covariates

We used a standardized questionnaire to collect personal information, including basic demographic characteristics (age, gender, marital status, education level, career, body mass index (BMI), and annual disposable income), habits of behavior (smoking and drinking), and personal chronic disease (including hypertension, malignant tumor, cataract, diabetes mellitus, hyperlipidemia, cholecystitis/gallstones, coronary heart disease, arthritis, heart disease, cerebral apoplexy, chronic gastrointestinal disease, hepatitis/cirrhosis, chronic bronchitis), which were conducted by public health workers and well-trained doctors. Anthropometric measurements including height, weight, and waist were taken by trained investigators using standard methods. BMI was classified as follows: (1) underweight, < 18.5 kg/m²; (2) normal weight, 18.5–23.9 kg/m²; (3) overweight, 24.0–27.9 kg/m²; and (4) obesity, ≥ 28.0 kg/m² (Zhong et al. 2021). Blood pressure was measured on each person's right arm three times with an electronic sphygmomanometer while sitting in a horizontal position. Marital status was classified

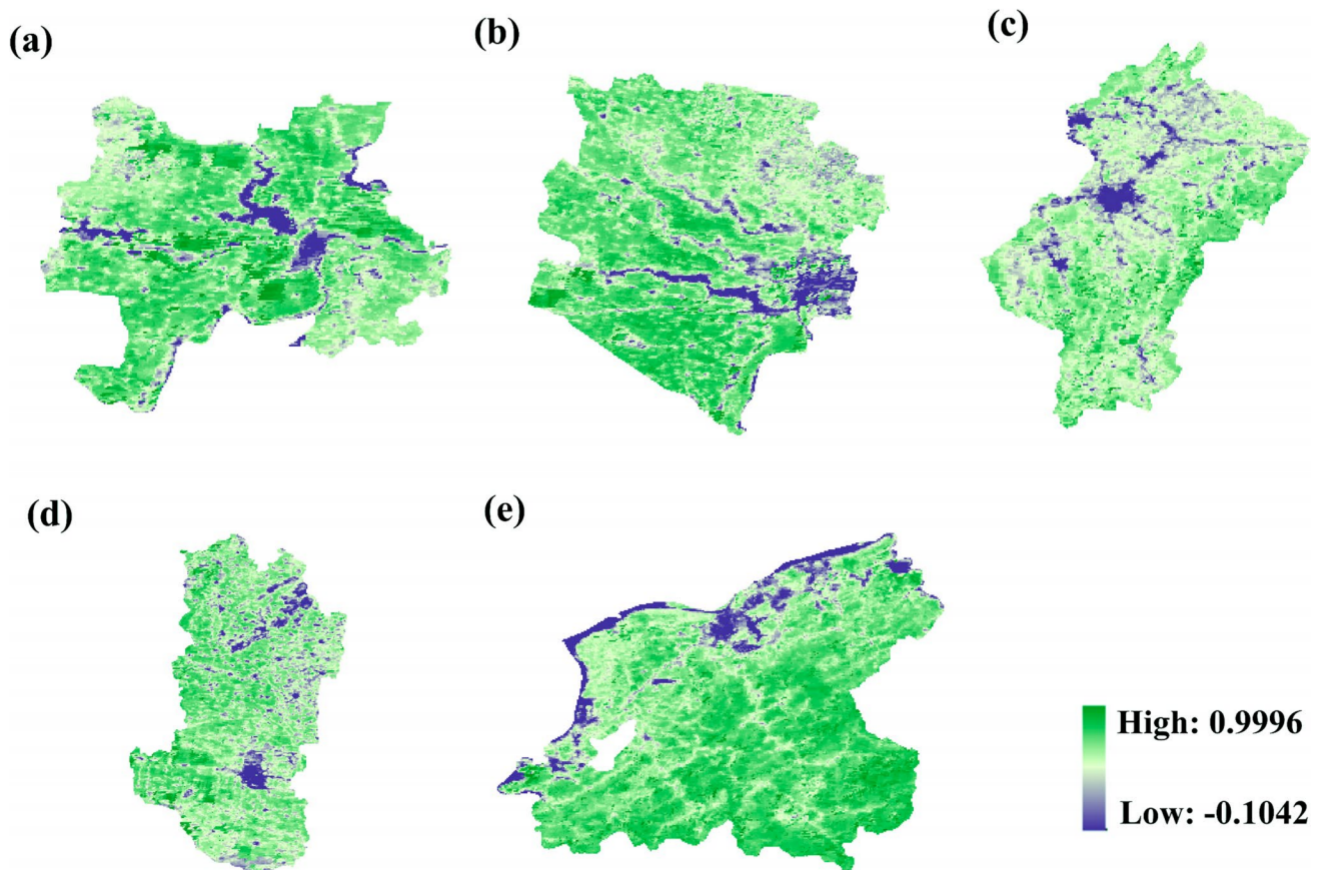


Fig. 1 NDVI density of five counties or districts in Anhui province in 2014. (a) is Wuhe county, (b) is Huaiyuan county, (c) is Qingyang county, (d) is Lingbi county, (e) is Guichi districts. Abbreviations: NDVI, normalized differential vegetation index

as married, widowed, divorced, and unmarried; career was classified as (1) agriculture, forestry, and fishery, (2) temporary workers, (3) housewives or husbands, (4) self-employed (including industrial and commercial services, freelance jobs), (5) regular workers (employees of enterprises and institutions, government civil servants, cadres mainly engaged in village management work), (6) unemployed, and (7) free work; education level was classified as elementary school or below, middle school, high school/ technical school, college, or above. The annual disposable income was below 1000 RMB, 1000–5000 RMB, 5000–10,000 RMB, 10,000 RMB, and above. Smoking status was divided into ever group and never group. Drinking status was divided into ever group and never group.

Statistical analyses

The normal and non-normal distributions of classified variables and continuous variables were represented by number (percentage), mean (standard deviation, SD), and median (quartile range), respectively. χ^2 test was used to compare the counting data. Generalized linear models were used to

evaluate the effects of green space coverage, carbon dioxide, and their interactions on sleep quality. First, we tested the individual associations between carbon dioxide or green space and sleep quality. Second, the additive effects of carbon dioxide and green space on sleep quality were then estimated simultaneously in the same model. Third, we explored the individual and interactional effects of carbon dioxide and green space on sleep quality and various items of sleep quality (including subjective sleep quality, time to fall asleep, sleep time, sleep efficiency, sleep disorders, hypnotic medications, and daytime dysfunction). Then, the model containing the interaction items is tested. Finally, we investigated the mediating effect of carbon dioxide, green space, and sleep quality. P value < 0.05 on both two sides indicated statistical significance. Interplot method was used to plot the interaction of carbon dioxide and green space on sleep quality. We adjusted all models for age, sex, marital status, the annual disposable income, education, smoking status, drinking status, personal chronic disease, mean temperature, and mean relative humidity. R software version 3.4.1 was used for data processing and analysis, and R-package of “interplot,” “splines,” and “mediation” was used to make

interaction plots, analysis of individual and additive effect, and analysis of mediating effect, respectively.

Results

Descriptive statistics

The prevalence of sleep disturbance in the overall elderly population was 58.40%. Table 1 shows that participants with

sleep disorders had a higher mean age (standard deviation, SD) than those with better sleep quality (71.0 (7.8) vs. 69.7 (7.3), $P < 0.001$). Except for annual disposable income and smoking status ($P = 0.305$ and 0.146), there were significant differences in the distribution of selected variables between individuals with sleep disorders and those with good sleep quality ($P < 0.05$). Table 2 shows a summary of PSQI score, CO₂, green space, and meteorological factors. The PSQI score (SD) of all participants was 6.85 (3.57). The mean values of CO₂ and NDVI were 3.40 and 0.82, respectively.

Table 1 Distributions of selected variables of the participants by with and without sleep disturbance

Variables	Good sleep	Sleep disturbance	<i>P</i> -value
Age (years, mean $\pm$ SD)	69.7 $\pm$ 7.3	71.0 $\pm$ 7.8	$< 0.001^a$
Gender (<i>n</i> , %)			$< 0.001^b$
Male	520 (47.3)	558 (36.2)	
Female	579 (52.7)	985 (63.8)	
Marital status (<i>n</i> , %)			$< 0.003^b$
Married	830 (75.5)	1064 (69.0)	
Widowed	243 (22.1)	431 (27.9)	
Divorced	3 (0.3)	5 (0.3)	
Unmarried	23 (2.1)	43 (2.8)	
Education level (<i>n</i> , %)			$< 0.001^b$
Elementary school or below	939 (40.3)	1392 (59.7)	
Middle school	120 (49.6)	122 (50.4)	
High school/ technical school or above	40 (58.0)	29 (42.0)	
Career (<i>n</i> , %)			$< 0.027^b$
Agriculture, forestry and fishery	967 (40.7)	1407 (59.3)	
Regular worker	54 (50.0)	54 (50.0)	
Other work	78 (48.8)	82 (51.4)	
The annual disposable income (RMB) (<i>n</i> , %)			0.305^b
< 1000	311 (39.8)	471 (60.2)	
1000~	472 (41.1)	677 (58.9)	
5000~	171 (44.1)	217 (55.9)	
> 10000	145 (44.9)	178 (55.1)	
Smoking status (<i>n</i> , %)			0.146^b
Never	691 (38.5)	1104 (61.5)	
Ever	405 (48.0)	438 (52.0)	
Drinking status (<i>n</i> , %)			$< 0.001^b$
Never	755 (39.4)	1161 (60.6)	
Ever	344 (47.4)	382 (52.6)	
Personal chronic disease (<i>n</i> , %)			$< 0.001^b$
Yes	363 (52.7)	326 (47.3)	
No	736 (37.7)	1217 (62.3)	

Abbreviations: SD standard deviation

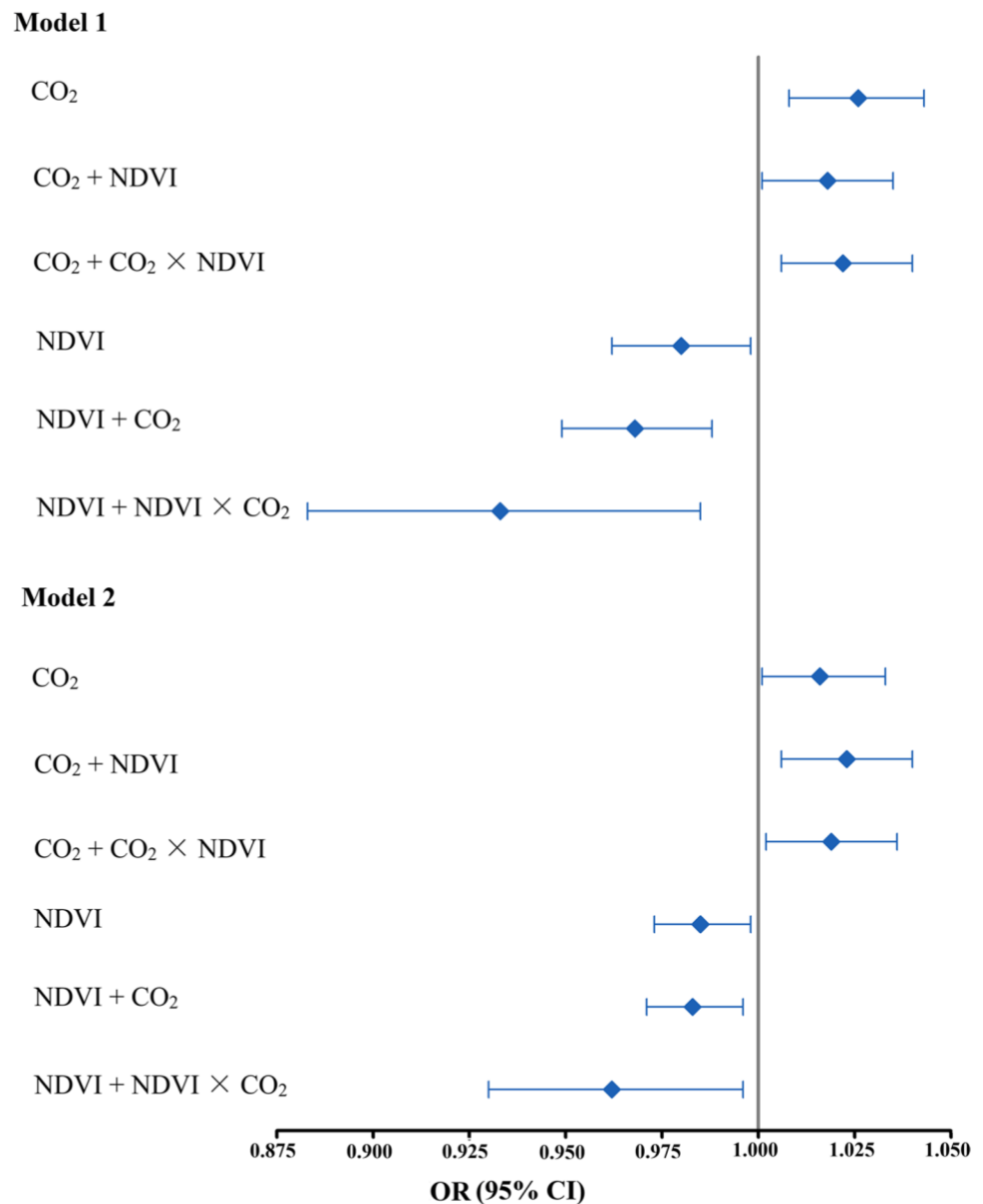
^aStudent's *t* test was used to compare normally distributed continuous variables between good sleep and sleep disturbance groups

^bA Chi-square test was used to test the distributions of categorical variables between good sleep and sleep disturbance groups

Regular worker contains employees of enterprises and institutions, government civil servants, and cadres mainly engaged in village management work; other work contains temporary workers, housewives or husbands, self-employed (including industrial and commercial services, freelance jobs), unemployed, and free work

	Min	P25	Median	Mean (SD)	P75	Max
CO ₂	1.84	2.17	2.97	3.40 (1.15)	4.53	5.00
NDVI	0.79	0.81	0.82	0.82 (0.02)	0.83	0.83
PSQI score	0.00	4.00	6.00	6.85 (3.57)	9.00	20.00
MT	8.78	15.30	15.49	15.30 (2.11)	15.89	17.18
MRH	71.17	71.17	75.94	75.58 (3.10)	77.76	79.03

Fig. 2 Estimated effect of CO₂, green space (NDVI), and their interaction on sleep quality of the elderly. Abbreviations: OR, odd ratio; CI, confidence interval; CO₂, carbon dioxide; NDVI, normalized differential vegetation index. Model 1: adjusted for age, sex, marital status, career, education level, the annual disposable income, smoking status, drinking status, BMI, and personal chronic diseases. Model 2: adjusted for mean temperature and relative humidity based on Model 1



In addition, we found that green space only had a statistically significant negative effect on sleep efficiency, sleep disorders, and daytime dysfunction. With each 0.1 increase in NDVI, the risk of low sleep efficiency, sleep disorders, and daytime dysfunction of the elderly decreased by 3.8% (OR: 0.962, 95% CI: 0.931, 0.993), 4.6% (OR: 0.954, 95% CI: 0.940, 0.968), and 4.4% (OR: 0.956, 95% CI: 0.933, 0.981), respectively.

Discussion

This study found that the prevalence of sleep disorders among the elderly in rural areas of Anhui Province was 58.40%, which was much higher than the average level in China (15.0%) and Canada (24.0%) (Chaput et al 2018; Cao

et al 2017). In the univariate analysis, we found that CO₂ was positively correlated with sleep quality after adjusting for related confounders, while green space was negatively correlated with sleep. In the interaction analysis, the negative correlation between green space and sleep quality decreased with the increase concentrations of CO₂. In the mediating analysis, green space was a mediating variable between the association between CO₂ and sleep quality.

Independent effects of CO₂ and green space on sleep quality of the elderly

In a single model, there may be a positive association between CO₂ and sleep quality, and green space had a protective effect on sleep quality of the elderly, which were

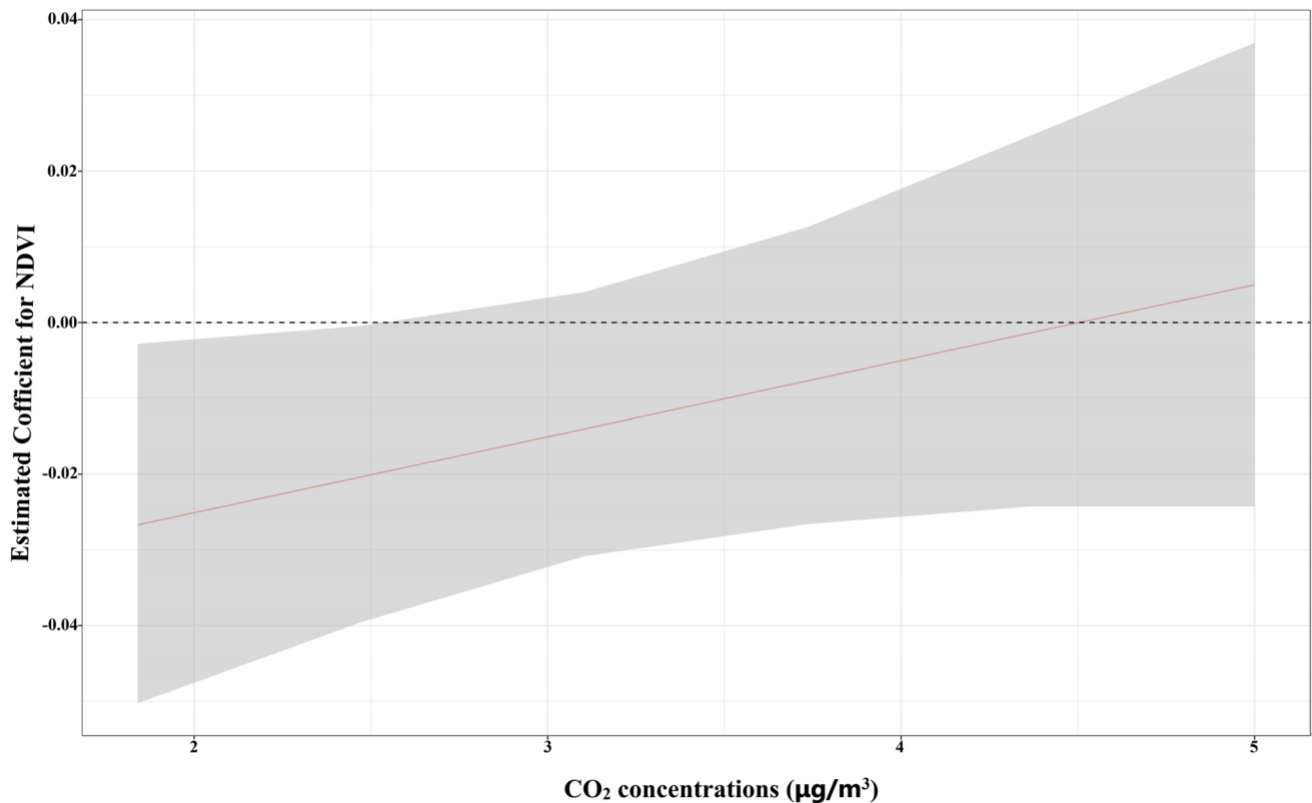


Fig. 3 Regression estimates of green space effects (95% confidence interval) on sleep quality of the elderly as a function of CO₂, by using interplot models, after adjusting for age, sex, marital status, career, education level, the annual disposable income, smoking status, drink-

ing status, BMI, and personal chronic diseases MT and MRH. Abbreviations: CO₂, carbon dioxide; NDVI, normalized differential vegetation index

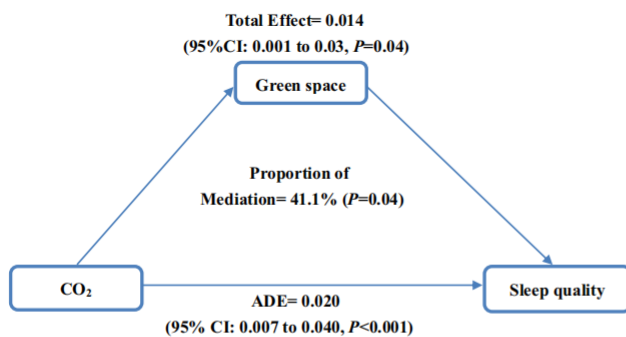


Fig. 4 The mediating effect of CO₂ and green space on sleep quality in the elderly. Abbreviations: ADE, average direct effects; CI, confidence interval; CO₂, carbon dioxide

consistent with some previous studies (Shin et al 2020; Zhang et al 2021). A Chinese experimental study showed a strong correlation between CO₂ and sleep quality (Xu et al 2021). Meanwhile, a Dutch study found that CO₂ had an effect on sleep efficiency and the number of wakes during the night (Mishra et al 2018). A review of green space found that adult residents with higher green space were less likely to suffer from sleep deprivation, and tree crowns were

associated with longer sleep duration (Billings et al 2020). At the same time, green space could promote walking and other healthy behaviors, reduce stress levels, improve mental health, and promote healthy sleep (Billings et al 2020). In addition, studies of the USA and Australia have also found that green space may have the protective effect on sleep quality (Astell-Burt et al 2013; Johnson et al 2018).

Although the biological mechanism of CO₂ and green space in sleep quality of the elderly were not well understood, some studies have elucidated the relationship. The current mainstream explanation for the mechanism of CO₂ acting on the respiratory system is as follows: when the human body absorbed CO₂ within a certain range, the pressure generated by physically dissolved CO₂ molecules in the arterial blood increased with the increase of the inhaled concentrations of CO₂, leading to the deepening and acceleration of respiration (Xu et al 2021; Adamovich et al 2019; Suyama et al 2003). When arterial carbon dioxide pressure rises to a certain level, it produced respiratory depression, reduced respiration, and decreased oxygen level of the body's cells. A range of physiological recovery mechanisms that should occur during sleep are blocked, resulting in poor sleep quality. Most older people

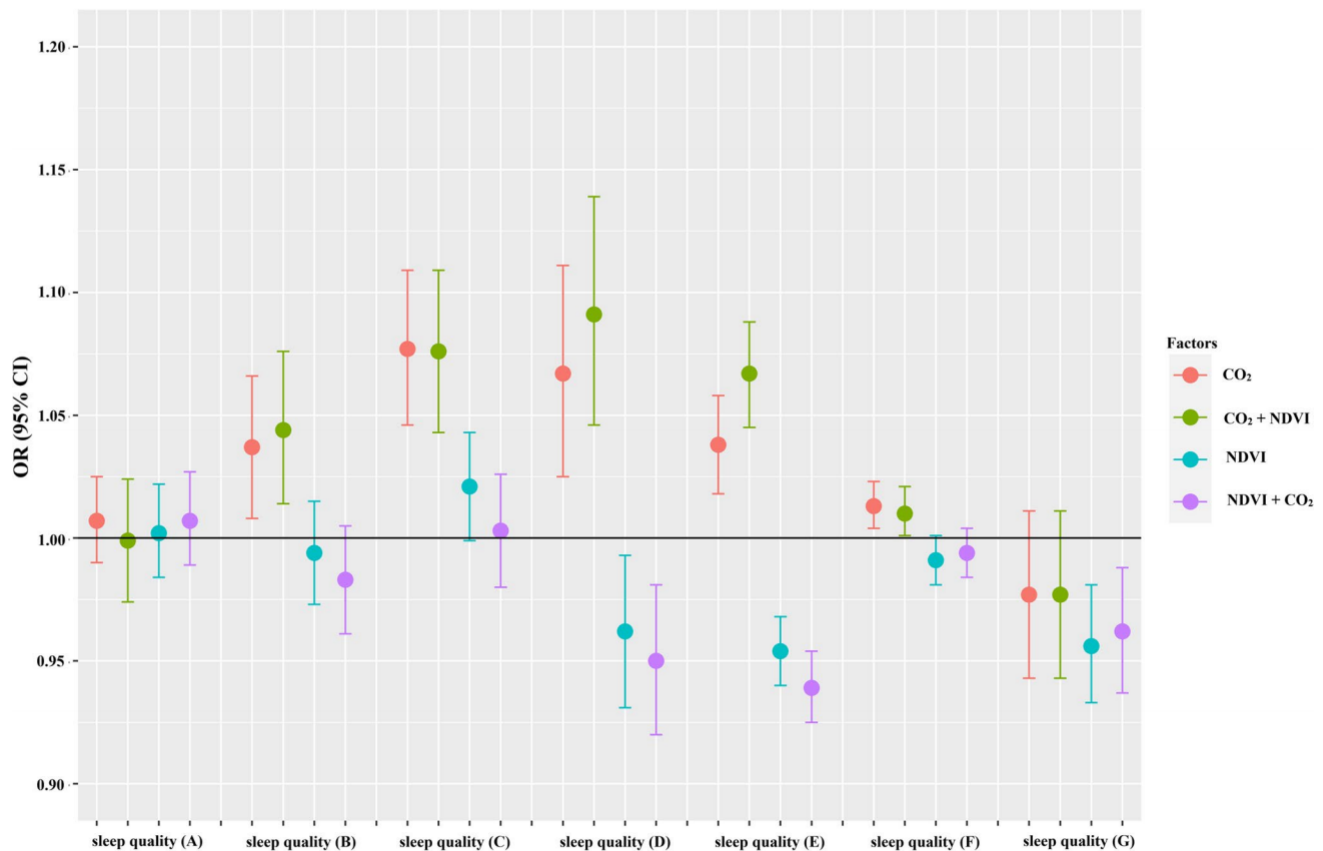


Fig. 5 The effect of CO₂ and green space on various items of sleep quality in the elderly. Various items of sleep quality were divided into: sleep quality (A), subjective sleep quality; sleep quality (B), time to fall asleep; sleep quality (C), sleep time; sleep quality (D),

sleep efficiency; sleep quality (E), sleep disorders; sleep quality (F), hypnotic medications; sleep quality (G), daytime dysfunction. Abbreviations: CI, confidence interval; CO₂, carbon dioxide; NDVI, normalized differential vegetation index

had chronic respiratory problems, so may be more vulnerable to CO₂.

There were also hypotheses about the mechanism between green space and sleep quality of the elderly. First, green space in residential areas could effectively prevent noise from industrial production and traffic, thus reducing the impact of noise on sleep at night or during the day (Johnson et al 2018). Second, some studies have showed that there was more exercise in areas with more green space coverage, and people improved their physical and mental health by increasing beneficial physical activity, which improved the quality of sleep (Danielsson et al 2013; Richardson et al 2013; Mytton et al 2012). Third, another mechanism by which green space affected sleep quality was through its relationship with social cohesion. Related studies have shown that areas with higher green space have higher social cohesion, which could reduce people's psychological burden, make people feel more secure, and promote good sleep quality (Johnson et al 2018; Hale et al 2013; Hill et al 2016).

Joint effects of CO₂ and green space on sleep quality of the elderly

We have found that the protective effect of green space on sleep quality of the elderly gradually decreased with the increase concentrations of CO₂ from the interactive plot of CO₂ and green space on sleep quality. At present, the mechanism of joint effects of CO₂ and green space on sleep quality of the elderly is unknown. After reviewing some plant literatures, we have proposed the following mechanism hypotheses for our research results: first of all, with the increase concentrations of CO₂, stomatal opening of plants may shrink and partially close, which reduced stomatal conductance, transpiration rate, plant volume, degree of stem, and leaf luxuriance (Hungate et al 1996), thus affecting the role of green space on sleep quality. Second, rising concentrations of CO₂ directly affected the climate through their radiative effects (trapping long-wave radiation) (Cao et al 2010). Plants used less water as temperatures rose. At the same time, the concentrations of O₃ on the surface became

larger and larger with the increase of temperature, and O_3 had the strong damage effect on most plants (Long et al 2006), and reduced the coverage of green space.

Mediation effects of CO_2 and green space on sleep quality of the elderly

The association between CO_2 and sleep quality of the elderly was mediated in part by increased green space, provided that the underlying hypothesis of mediating analysis was true. The proportion of mediation effects depended on the size of the surrounding green space and the concentrations of CO_2 . The choice of green space as a confounding or mediating factor in the association between CO_2 and sleep quality would influence the effect size one should use to assess the potential effects of CO_2 . If green space was primarily a confounding factor, the double-exposure model in Fig. 2 (corrected for green space as a confounding variable) should be used to capture the true effect of CO_2 on sleep. Otherwise, the single exposure model should be used to obtain the corresponding effect size, and we predicted that about 41.1% of the association between CO_2 and sleep quality may be related to increased green space. Higher green space coverage could limit dispersion by setting up green barriers or removing CO_2 , thus reducing concentrations of CO_2 near residential areas. These mechanisms could partially explain the empirical correlation between green rate and CO_2 . What's more, in green areas, residents tended to have lower CO_2 exposure due to less CO_2 sources for photosynthesis by green plants (Korner 2006). This may reflect a common source or lack thereof and therefore was not simply a causal pathway from CO_2 to sleep quality. In addition, green space may affect sleep of the elderly in a number of ways. For example, larger green space could stimulate physical activity (and thus reduce overweight and obesity) and reduce stress (Klompmaaker et al 2019). In our mediating analysis, we adjusted for chronic disease, weight status, and lifestyle habits (smoking and drinking). However, it was estimated that the direct effect of CO_2 on sleep quality remained.

Strengths and limitations

Strengths of this study include the following: first, the questionnaire survey and physical examination in this study were conducted by professional clinicians and public health workers, and the data had great integrity and accuracy. Second, in addition to exploring the independent effects of CO_2 and green space on sleep, we were able to investigate the confounding, interaction, and mediation effects of common chronic diseases, meteorological factors, and the several environmental exposures. Third, this study focused on the elderly with more vulnerable immune levels and sleep

quality, filling in the gaps of environmental factors and sleep in the older population.

There were still some limitations in our study: first, this study was a cross-sectional study, and the ability to make causal inferences was not very strong. Our subsequent prospective study would further validate the specific causal inference. Second, traffic noise and night light, as important indicators affecting sleep, were not included in the analysis of this study. Our study covered five rural areas in Anhui Province, where transportation and nighttime lighting facilities were relatively backward compared to some large cities, so the effect on sleep was minimal. Last, we took the annual mean concentrations of CO_2 and the NDVI in each county or district where the individuals lived as the individuals' exposure, which may overestimate or underestimate the actual exposure dose of the individuals. Because the local climatic and geographical conditions of each district or county that we investigated in Anhui Province were very similar, and the heterogeneity of individuals' exposure to CO_2 and green space in each district or county was very low, such overestimation or underestimation had little difference with the real value.

Conclusions

This study showed that CO_2 was positively correlated with sleep quality of the elderly in rural areas of Anhui Province, while the negative association was found between green space and sleep quality. Meanwhile, the correlation between green space and sleep quality decreased with concentrations of CO_2 increased. And the harmful effects of CO_2 on sleep quality of the elderly would decrease with the increase of green space. It was suggested that relevant departments should take some measures to increase the coverage of green plants and control the level of CO_2 emissions in rural areas, so as to reduce the burden of diseases caused by sleep quality of the elderly.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11356-021-17296-7>.

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Author contribution Guoao Li and Fen Huang collaboratively designed the study, both making substantial intellectual contributions. Guoao Li analyzed the data and drafted the manuscript together with Zhenyu Zhu, and Mingjun Hu and Fen Huang revised the manuscript. Jialiu He and Wanjun Yang contributed in collecting the data. Jinliang Zhu, Huanhuan Zhao, and Hanshuang Zhang all contributed in interpreting the data. Guoao Li and Zhenyu Zhu contributed equally. All authors read and approved the final manuscript.

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Data availability The datasets analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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